

EXPERIMENT NO: 05

Data Collection and Initial Exploration

PROGRAM:-

1. Data Collection:

```
import pandas as pd

db=pd.read_csv("diabetess.csv")

print(db)
```

OUTPUT:-

```
   Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin   BMI \
0             6     148             72             35         0  33.6
1             1      85             66             29         0  26.6
2             8     183             64              0         0  23.3
3             1      89             66             23        94  28.1
4             0     137             40             35       168  43.1
..          ...     ...             ...             ...     ...   ...
763           10     101             76             48       180  32.9
764            2     122             70             27         0  36.8
765            5     121             72             23       112  26.2
766            1     126             60              0         0  30.1
767            1      93             70             31         0  30.4

   DiabetesPedigreeFunction  Age  Outcome
0                0.627    50         1
1                0.351    31         0
2                0.672    32         1
3                0.167    21         0
4                2.288    33         1
..                 ...     ...       ...
763               0.171    63         0
764               0.340    27         0
765               0.245    30         0
766               0.349    47         1
767               0.315    23         0

[768 rows x 9 columns]
```

2.Initial Exploration

```
print(db.info())

print(db.describe())

import matplotlib.pyplot as plt

import seaborn as sns

db.hist(bins=50,figsize=(20,15))

plt.show()

sns.pairplot(db)

plt.show()
```

OUTPUT:-

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 768 entries, 0 to 767
Data columns (total 9 columns):
#   Column                Non-Null Count  Dtype  
---  -
0   Pregnancies            768 non-null    int64  
1   Glucose                768 non-null    int64  
2   BloodPressure          768 non-null    int64  
3   SkinThickness          768 non-null    int64  
4   Insulin                768 non-null    int64  
5   BMI                   768 non-null    float64 
6   DiabetesPedigreeFunction 768 non-null    float64 
7   Age                   768 non-null    int64  
8   Outcome               768 non-null    int64  
dtypes: float64(2), int64(7)
memory usage: 54.1 KB
None
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin
count	768.000000	768.000000	768.000000	768.000000	768.000000
mean	3.845052	120.894531	69.105469	20.536458	79.799479
std	3.369578	31.972618	19.355807	15.952218	115.244002
min	0.000000	0.000000	0.000000	0.000000	0.000000
25%	1.000000	99.000000	62.000000	0.000000	0.000000
50%	3.000000	117.000000	72.000000	23.000000	30.500000
75%	6.000000	140.250000	80.000000	32.000000	127.250000
max	17.000000	199.000000	122.000000	99.000000	846.000000

	BMI	DiabetesPedigreeFunction	Age	Outcome
count	768.000000	768.000000	768.000000	768.000000
mean	31.992578	0.471876	33.240885	0.348958
std	7.884160	0.331329	11.760232	0.476951
min	0.000000	0.078000	21.000000	0.000000
25%	27.300000	0.243750	24.000000	0.000000
50%	32.000000	0.372500	29.000000	0.000000
75%	36.600000	0.626250	41.000000	1.000000
max	67.100000	2.420000	81.000000	1.000000

